Alaa Abdel Fattah Allam

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Master's researcher in Artificial Intelligence & Data Science specializing in Foundation Model architectures, large-scale data curation pipelines, and model evaluation & safety auditing. Hands-on expertise in modern Transformer design (RoPE, Grouped-Query Attention, SwiGLU, Sparse MoE), high-throughput data processing (MinHash LSH deduplication, custom BPE tokenization), distributed deep learning dynamics (FSDP, ZeRO-3, MFU optimization), and counterfactual bias/hallucination risk analysis. Proven empirical research background at the National Research Centre.

EDUCATION

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Cairo, Egypt | May 2025 – Ongoing

Master of Science (M.Sc.) in Artificial Intelligence – Data Science Major

- **Research Focus:** Foundation Model Pre-training, Attention Kernels (GQA/MLA), Sparse Mixture-of-Experts (MoE), Data Deduplication & Curation Pipelines, Empirical Safety Alignment, and Distributed Deep Learning.

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Alamein, Egypt | Sept 2020 – July 2024

Bachelor of Science (B.Sc.) in Artificial Intelligence – Data Science Major (GPA: 3.5 / 4.0, Ranked 3rd in Department)

- **Foundational Coursework:** Deep Learning, Advanced NLP, Computer Vision, Distributed Databases, Data Structures & Algorithms, Mathematical Optimization, Multivariable Calculus, Linear Algebra, Probability & Applied Statistics.
- **Graduation Research Project:** Deep neural network architecture for document intelligence and generative data augmentation (Presented at the 1st International Conference on Universities and Artificial Intelligence Technology, Amman, Jordan).

TECHNICAL SKILLS

Foundation Model Architectures: Decoder-Only Transformers, Grouped-Query Attention (GQA), Rotary Position Embeddings (RoPE), SwiGLU FFN, Sparse Mixture of Experts (MoE with Top-K Routing & Router Z-Loss), RMSNorm, KV Caching, FlashAttention.

Data Curation & Processing: Large-Scale Pretraining Data Pipelines, MinHash & LSH Deduplication, Heuristic Quality & Perplexity Filtering (RefinedWeb/FineWeb criteria), Custom BPE & SentencePiece Tokenizer Training, Memory-Mapped Datasets (.bin/.idx, Arrow, Parquet).

Evaluation, Safety & Risk: LM Evaluation Harness, Cross-Entropy & Perplexity Benchmarking, Multi-Choice Reasoning (MMLU/ARC log-likelihood scoring), Counterfactual Demographic Bias Probing, Hallucination Entropy Auditing, Red-Teaming, Alignment.

Distributed HPC & Systems: PyTorch, Fully Sharded Data Parallel (FSDP), DeepSpeed (ZeRO-1/2/3), Mixed Precision (BF16/FP8), Gradient Norm Tracking, Cosine Annealing Schedulers, Model FLOPs Utilization (MFU), Weights & Biases, Docker, Linux/Bash.

Programming & Math: Python (NumPy, SciPy, PyTorch, Pandas), C, SQL, Mathematical Optimization, Information Theory, High-Dimensional Statistics.

RESEARCH & ACADEMIC EXPERIENCE

National Research Centre (NRC) — Giza, Egypt | Jan 2025 – Present

AI Research Consultant

- Formulate empirical deep learning methodologies and guide interdisciplinary research scientists in adapting neural architectures and predictive models to complex, high-dimensional scientific datasets.
- Architect automated data curation, sanitization, and feature validation pipelines to eliminate data contamination, feature leakage, and measurement bias across experimental domains.
- Design reproducible experimental baselines and ablation studies, documenting model convergence dynamics, loss trajectories, and statistical significance tests to support scientific publication.

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Cairo & Alamein, Egypt | Sept 2024 – Present

Graduate Teaching Assistant – Artificial Intelligence & Computer Science

- Instruct and mentor 90+ undergraduate students in Python, Data Structures & Algorithms, Database Systems, and Machine Learning foundations.
- Design computational problem sets and lab assignments; conduct rigorous code reviews emphasizing algorithmic complexity ($O(N)$ bounds), memory layouts, and clean modular software architecture.

Technocolabs Softwares — Remote | Sept 2023 – Nov 2023

Machine Learning & Data Science Research Intern

- Researched and benchmarked supervised learning classification models and NLP text sentiment algorithms on noisy, real-world data.
- Executed exploratory data analysis (EDA), automated outlier imputation, and cross-validation workflows to evaluate generalization.

SELECTED RESEARCH & ENGINEERING PROJECTS

OpenFoundation-Engine: Modular LLM Pretraining, Data Curation & Safety Evaluation Framework — github.com/alaa01-bit/OpenFoundation-Engine

Python, PyTorch / NumPy, FastAPI, MinHash LSH, Transformers, RoPE, MoE, Distributed HPC

- **Architecture & Kernels:** Implemented a modular decoder-only Transformer incorporating Grouped-Query Attention (4:1 GQA ratio slashing KV cache memory by 75%), Rotary Position Embeddings (RoPE), SwiGLU activations, and Sparse Mixture of Experts (MoE) with Top-2 routing and auxiliary load-balancing loss.
- **Data Curation & Tokenization:** Built an end-to-end pretraining data curation pipeline with 64-permutation MinHash LSH deduplication, heuristic quality/repetition filtering, and a custom BPE tokenizer engine with compression analytics (1.62× compression ratio).
- **Evaluation & Safety Harness:** Developed an automated evaluation suite tracking validation perplexity, MMLU multi-choice log-likelihood scoring, and counterfactual log-odds demographic bias probing ($\Delta \log P$) alongside token prediction entropy for hallucination auditing.
- **Distributed HPC Planning:** Formulated distributed VRAM capacity models simulating per-GPU memory footprints across ZeRO-1/2/3 and PyTorch FSDP, benchmarking hardware Model FLOPs Utilization (MFU) on high-performance GPU clusters.

Document Intelligence & Generative Data Augmentation Framework — github.com/alaa01-bit/Document-Intelligence

Deep Learning, PyTorch, Computer Vision, GANs (DCGAN, StyleGAN2-ADA), Jordan AI Conference

- Developed an end-to-end OCR and document verification framework; presented findings at the 1st Int. Conference on Universities and AI Technology.
- Trained a deep neural network for Arabic digit recognition reaching **99.5% accuracy**; evaluated generative adversarial augmentation (DCGAN, StyleGAN2) to synthesize hard negative document samples, improving out-of-distribution robustness.

Autonomous Multi-Agent Topology for Code Reasoning & Alignment — github.com/alaa01-bit/Agentic-Reasoning

Hierarchical Agent Orchestration, Static Analysis, Safety Alignment

- Architected an autonomous multi-agent reasoning system evaluating complex code for architectural vulnerabilities, performance bottlenecks, and algorithmic efficiency; analyzed agent reasoning traces, failure modes, and cyclic dependencies.